

1. Calculation method: Topology-Weighted

	Enriched gained GO terms	P-value	Enriched lost GO terms	P-value
Unikonta	- protein import into peroxisome matrix, docking - cAMP catabolic process	9.5E-3 1.9E-2		
Opisthokonta	- regulation of transcription - regulation of nucleobase, nucleoside, nucleotide and nucleic acid metabolic process	2.3E-2 2.8E-2	- protein-heme linkage - asparagine biosynthetic process	5.2E-3 1.0E-2
Metazoa & Choanoflagellata	- cell communication - galactosylceramide catabolic process	3.6E-6 9.2E-3	- xylan catabolic process - hemicellulose metabolic process - polysaccharide catabolic process *	1.6E-5 1.5E-4 3.2E-4
Metazoa	- regulation of transcription, DNA-dependent - transcription, DNA-dependent	1.2E-7 1.8E-6	- aromatic amino acid family biosynthetic process, prephenate pathway - histidine biosynthetic process	1.1E-4 2.3E-3
Bilaterian & Cnidaria	- apoptosis - peptide cross-linking	3.1E-4 4.7E-4	- protein folding - transcription initiation	1.7E-3 3.8E-3
Bilateria	- mitochondrial electron transport, NADH to ubiquinone - mitochondrial ATP synthesis coupled electron transport	8.3E-6 1.1E-5	- branched chain family amino acid biosynthetic process - cellular amino acid biosynthetic process - water-soluble vitamin biosynthetic process *	3.3E-4 6.3E-4 5.0E-3
Deuterostomia	- protein transport - protein localization	8.1E-2 8.6E-2	- dicarboxylic acid metabolic process - phosphoenolpyruvate-dependent sugar phosphotransferase system	1.6E-3 3.2E-3
Chordata	- feeding behavior - G-protein coupled receptor protein signaling pathway	4.3E-5 7.0E-4	proteolysis	2.1E-2
Urochordata & Vertebrata	- antigen processing and presentation - protein amino acid phosphorylation	5.5E-3 1.8E-2	- folic acid and derivative metabolic process - group transfer coenzyme metabolic process	2.3E-3 5.8E-3
Vertebrata	- immune response - cell surface receptor linked signal transduction	4.4E-11 2.0E-6	DNA topological change	2.0E-3
Tetrapoda	- regulation of growth - synaptic transmission	1.3E-2 2.0E-2	- valyl-tRNA aminoacylation - response to water	4.3E-3 8.6E-3
Amniota	- immune response - defense response	1.8E-3 2.0E-3	- regulation of transcription, DNA-dependent - protein secretion - riboflavin biosynthetic process * - thiamin biosynthetic process *	9.2E-8 5.1E-5 1.0E-3 1.8E-3
Mammalia	- hemopoiesis - reciprocal meiotic recombination	2.8E-3 8.3E-3	- aromatic amino acid family biosynthetic process - chorismate metabolic process	1.1E-2 1.3E-2

2. Calculation method: Parent-Child-Union, multiple testing correction (MTC): Bonferroni

	Enriched gained GO terms	P-value (adjusted)	Enriched lost GO terms	P-value (adjusted)
Unikonta				
Opisthokonta				
Metazoa & Choanoflagellata	- cell communication - biological regulation	1.3E-3 8.3E-2	- carbohydrate metabolic process - polysaccharide metabolic process	3.5E-4 3.0E-2
Metazoa	- regulation of cellular process - regulation of nitrogen compound metabolic process	6.4E-15 6.5E-6	organic acid biosynthetic process	9.1E-2
Bilaterian & Cnidaria	regulation of biosynthetic process	3.6E-2		
Bilateria	- electron transport chain - phosphorus metabolic process - regulation of cellular process *	2.0E-2 3.4E-2 4.1E-2		
Deuterostomia			- organic acid metabolic process - cellular amino acid and derivative metabolic process	4.7E-3 2.7E-2
Chordata				
Urochordata & Vertebrata		5.5E-3 1.8E-2	carbohydrate metabolic process	3.5E-2
Vertebrata	- immune system process - regulation of cellular process	3.8E-8 4.9E-8	carbohydrate metabolic process	3.3E-2
Tetrapoda				
Amniota	- response to stimulus - immune system process	1.9E-2 8.3E-2	- regulation of macromolecule metabolic process - regulation of gene expression	2.9E-3 1.4E-2
Mammalia				

3. Calculation method: Parent-Child-Intersection, multiple testing correction (MTC): Bonferroni

	Enriched gained GO terms	P-value (adjusted)	Enriched lost GO terms	P-value (adjusted)
Unikonta				
Opisthokonta				
Metazoa & Choanoflagellata	- cell communication - biological regulation	1.0E-3 6.3E-2	carbohydrate metabolic process	2.6E-4
Metazoa	biological regulation	4.3E-13		
Bilaterian & Cnidaria				
Bilateria	- phosphorus metabolic process - biological regulation	2.4E-2 3.4E-2		
Deuterostomia			- organic acid metabolic process - cellular amino acid and derivative metabolic process	3.3E-3 8.0E-3
Chordata				
Urochordata & Vertebrata		5.5E-3 1.8E-2	- cellular nitrogen compound metabolic process - carbohydrate metabolic process	2.2E-2 2.7E-2
Vertebrata	- immune system process - biological regulation	2.2E-8 1.7E-7	carbohydrate metabolic process	2.6E-2
Tetrapoda				
Amniota	- response to stimulus - immune system process	1.6E-2 7.2E-2		
Mammalia				

4. Calculation method: Term-For-Term, multiple testing correction (MTC): Bonferroni

	Enriched gained GO terms	P-value (adjusted)	Enriched lost GO terms	P-value (adjusted)
Unikonta				
Opisthokonta				
Metazoa & Choanoflagellata	- cell communication - signal transduction	1.0E-3 3.4E-2	- carbohydrate metabolic process - xylan metabolic process	1.5E-4 7.6E-3
Metazoa	- regulation of cellular process - regulation of biological process - regulation of RNA metabolic process *	5.2E-11 3.0E-10 5.1E-5	aromatic amino acid family biosynthetic process, prephenate pathway	2.6E-2
Bilaterian & Cnidaria				
Bilateria	- mitochondrial electron transport, NADH to ubiquinone - ATP synthesis coupled electron transport	3.7E-3 6.1E-2	organic acid biosynthetic process	5.1E-2
Deuterostomia			- organic acid metabolic process - cellular amino acid and derivative metabolic process	4.8E-3 9.9E-3
Chordata	- feeding behavior - digestion	9.1E-3 2.7E-2		
Urochordata & Vertebrata		5.5E-3 1.8E-2	carbohydrate metabolic process	3.5E-2
Vertebrata	- immune response - biological regulation	2.2E-8 9.0E-6	carbohydrate metabolic process	1.3E-2
Tetrapoda				
Amniota	- response to stimulus - immune response	4.6E-2 9.1E-2	- regulation of RNA metabolic process - transcription, DNA-dependent	4.9E-5 2.5E-4
Mammalia	system development	2.7E-2		

5. Calculation method: Term-For-Term, multiple testing correction (MTC): Westfall-Young-Single-Step

	Enriched gained GO terms	P-value (adjusted)	Enriched lost GO terms	P-value (adjusted)
Unikonta				
Opisthokonta				
Metazoa & Choanoflagellata	- cell communication - signal transduction	2.0E-3 4.2E-2	- carbohydrate metabolic process - xylan metabolic process	0 6.0E-3
Metazoa	- biological regulation - transcription - signal transduction *	0 3.8E-2 4.6E-2	aromatic amino acid family biosynthetic process, prephenate pathway	6.4E-2
Bilaterian & Cnidaria	programmed cell death	7.6E-2		
Bilateria	- mitochondrial electron transport, NADH to ubiquinone - ATP synthesis coupled electron transport	2.0E-3 3.2E-2	organic acid biosynthetic process	6.8E-2
Deuterostomia			- organic acid metabolic process - cellular amino acid and derivative metabolic process	4.0E-3 8.0E-3
Chordata	- feeding behavior - digestion	2.0E-2 5.8E-2		
Urochordata & Vertebrata				
Vertebrata	- biological regulation - cell surface receptor linked signal transduction - antigen processing and presentation	0 2.0E-3 6.0E-3	carbohydrate metabolic process	2.0E-2
Tetrapoda				
Amniota			- regulation of RNA metabolic process - regulation of macromolecule metabolic process	0 2.0E-3
Mammalia	system development	4.4E-2		

6. Calculation method: Term-For-Term, multiple testing correction (MTC): Westfall-Young-Step-Down

	Enriched gained GO terms	P-value (adjusted)	Enriched lost GO terms	P-value (adjusted)
Unikonta				
Opisthokonta				
Metazoa & Choanoflagellata	- cell communication - signal transduction	0 3.6E-2	- carbohydrate metabolic process - xylan metabolic process	1.0E-3 8.0E-3
Metazoa	- regulation of cellular process - transcription - signal transduction *	0 3.9E-2 5.4E-2	aromatic amino acid family biosynthetic process, prephenate pathway	4.8E-2
Bilaterian & Cnidaria	programmed cell death	7.1E-2		
Bilateria	- mitochondrial electron transport, NADH to ubiquinone - ATP synthesis coupled electron transport	1.0E-3 2.7E-2	- organic acid metabolic process - organic acid biosynthetic process	1.0E-3 5.0E-3
Deuterostomia			- organic acid metabolic process - cellular amino acid and derivative metabolic process	4.8E-3 9.9E-3
Chordata	- feeding behavior - digestion	2.8E-2 6.1E-2		
Urochordata & Vertebrata				
Vertebrata	- biological regulation - multicellular organismal process - antigen processing and presentation *	1.0E-3 5.0E-3 9.0E-3	carbohydrate metabolic process	3.2E-2
Tetrapoda				
Amniota	response to stimulus	9.9E-2	- regulation of RNA metabolic process - RNA biosynthetic process	0 1.0E-3
Mammalia	system development	5.5E-2		